

SA-UAED: Joint Frame-Level Detection of Audio Events, Speaker Activities, and Speaker-Attributed Paralinguistic Events

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Abstract

Existing sound event detection (SED) and speaker diarization (SD) systems typically fail to attribute paralinguistic events (e.g., laughter, coughing) to individual speakers, primarily due to the scarcity of fine-grained labeled data. To address this, we propose a dedicated simulation pipeline generating large-scale audio mixtures with precise frame-level annotations. Leveraging this, we introduce the Speaker-Attributed Unified Audio Event Detection (SA-UAED) framework for joint frame-level prediction of sound events, speaker activities, and paralinguistics. Experiments demonstrate that SA-UAED substantially improves speaker-attributed paralinguistic detection without compromising generic SED and SD accuracy.

Index Terms: Speaker diarization, paralinguistic event detection, sound event detection, audio simulation

1. Introduction

Real-world acoustic environments are highly complex, often comprising simultaneous multi-speaker speech, paralinguistic events (such as laughter and coughing), and diverse background sound events. These overlapping acoustic phenomena are ubiquitous in everyday scenarios, including family homes, open-plan offices, and clinical monitoring rooms. For instance, a spoken dialogue system must distinguish a user’s laughter from background music and another person’s cough to provide appropriate responses, while healthcare systems rely on accurate detection of coughing patterns alongside speech to monitor respiratory conditions remotely. Accurate, frame-level detection and separation of these heterogeneous events are therefore essential for enabling robust, context-aware audio applications such as intelligent meeting transcription with emotional cues, multi-party dialogue systems, and real-time patient surveillance.

Driven by the rapid development of speech foundation models, fine-grained paralinguistic annotations and understanding of speech data have also attracted increasing research attention [1, 2]. While existing sound event detection (SED) [3, 4, 5] and speaker diarization (SD) [6, 7, 8, 9] techniques can detect general audio events and speaker occurrences, they often cannot associate paralinguistic events (e.g., laughter, cough, etc.) with individual speakers [10]. Recent efforts toward a unified framework, most notably the Transformer-based Unified Audio Event Detection (T-UAED) [10] model, have demonstrated the feasibility of jointly modeling non-speech sound events and speaker-aware speech activities by conditioning on speaker embeddings. However, T-UAED still suffers from two critical shortcomings. First, it completely ignores paralinguistic events that frequently co-occur with speech. This is partially caused by the lack of sufficient speech data with diverse speakers and fine-grained

speaker-attributed paralinguistic labels. Second, it relies on generic speaker embeddings for speaker distinction, which are sub-optimal for capturing the short-duration and highly variable spectral and temporal characteristics of paralinguistic events. In particular, our preliminary experiments suggest that existing speaker embeddings fail to reliably distinguish the laughter and cough of different speakers, despite their strong performance on speaker recognition benchmarks.

In this paper, we aim to address the aforementioned limitations in both data and modeling aspects, ultimately building a unified audio event detection (UAED) system capable of detecting speaker-attributed paralinguistic events, speech activities, and general audio events. Our contributions can be summarized as follows:

1. We propose a dedicated audio simulation pipeline to accommodate the development of UAED systems with paralinguistic modeling. This simulation pipeline overcomes the prohibitive cost and annotation challenges of real-world multi-label data while providing controlled, perfectly aligned ground truth.
2. Exploiting the proposed data simulation pipeline, we introduce LibriPara, a large-scale 500-hour simulated dataset of 16 kHz audio featuring multi-speaker speech and paralinguistics, along with frame-level event annotations.
3. We present SA-UAED (Speaker-Attributed Unified Audio Event Detection), a novel framework for joint frame-level detection of general audio events, speaker activities, and paralinguistic events. It is tailored for associating paralinguistic events with speaker identities, requiring minimal modifications to existing UAED models without compromising SED and SD performance.

2. Related Work

2.1. Sound Event Detection and Acoustic Representations

Sound Event Detection (SED) has evolved from foundational CNNs (e.g. PANNs [3]) to powerful Transformer-based architectures like BEATs [11] and Whisper [12] variants. While these models excel at capturing diverse environmental sounds, standalone SED systems typically lack the speaker-aware context necessary to disentangle targeted human vocalizations from complex, overlapping multi-talker backgrounds.

2.2. Speaker Diarization and Speaker-Attributed Paralinguistic Detection

Modern speaker diarization relies on robust embeddings (e.g. x-vectors [13] or ECAPA-TDNN [14]) explicitly optimized to capture articulated speech traits while deliberately suppress-

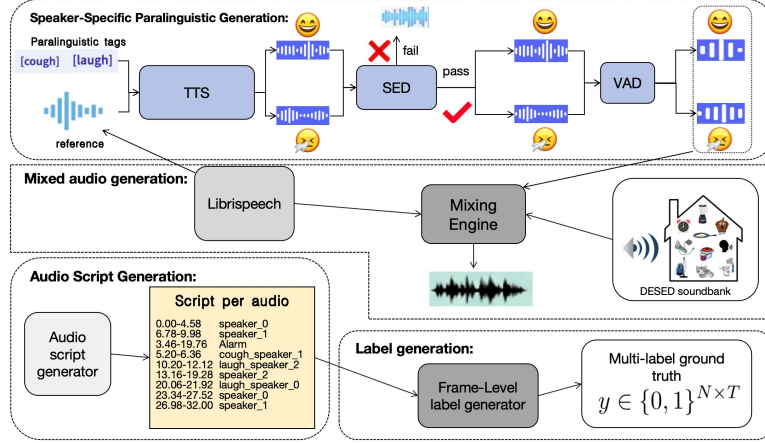


Figure 1: *Proposed audio simulation pipeline (pure paralinguistic generation using ChatterBox-Turbo with LibriSpeech references, PANNs filtering, VAD trimming and mixing with DESED events).*

ing environmental and non-speech variations. Conversely, Paralinguistic Event Detection (PED) identifies the occurrence of cues like laughter and coughing but ignores the speaker identity (“who”). Unifying these tasks is severely hindered by an acoustic mismatch: transient paralinguistic bursts (e.g. erratic glottal variations in laughter or explosive wideband energy in coughing) drastically alter typical vocal tract configurations. Consequently, standard speech-biased embeddings fail to maintain identity consistency during such events.

2.3. Unified Audio Understanding Frameworks

To bridge the gap between speech and non-speech analysis, unified frameworks like the Transformer-based Unified Audio Event Detection (T-UAED) [10] model elegantly enable joint SED and diarization by conditioning a shared architecture on pre-extracted speaker embeddings. However, they harbor a critical blind spot: the assumption that a generic speech-derived embedding can generalize to all human vocalizations. Because existing models route non-verbal vocalizations through mismatched, speech-biased embedding pipelines, they fail to robustly disentangle speaker identities during paralinguistic events. This architectural gap necessitates the task-specific representation decoupling strategy proposed in our work.

3. Methodology

3.1. Data Simulation Strategies

To address the severe scarcity of multi-label audio data featuring overlapping speech and paralinguistics, we design a highly-controlled audio simulation pipeline leveraging state-of-the-art zero-shot text-to-speech (TTS) systems. As shown in Figure 1, it consists of four phases:

(1) Audio Script Generation: As the foundational step of our simulation pipeline, we employ a stochastic audio script generator to orchestrate the complex temporal dynamics of the acoustic scene. To facilitate fine-grained labeling of each sample, this generator synthesizes a deterministic, text-based recipe (as illustrated in the lower-left corner of Figure 1) prior to waveform rendering. This script meticulously defines the precise onset and offset timestamps, event categories (e.g. speech, other vocal sounds, and environmental events), and explicit speaker identities for every vocal event. The audio script generator

can be configured to simulate different speech overlap ratios, speaker counts, occurrences of paralinguistic and general audio events, ensuring controllability and density. Crucially, this pre-defined timeline also greatly reduces the effort needed for fine-grained data labeling.

(2) Speaker-Specific Paralinguistic Generation: The core challenge in paralinguistic diarization is ensuring consistency between a speaker’s articulated speech and their spontaneous non-verbal vocalizations. To achieve this, we employ an advanced zero-shot TTS model, ChatterBox-Turbo [15], for speaker-specific paralinguistic sound generation. In this paper, we consider *laughter* and *cough* as two representative paralinguistic events, which can be directly generated by the ChatterBox-Turbo model via special tokens (i.e., [cough] and [laugh]) in the transcript. We take the audio clips of each speaker in the LibriSpeech [16] corpus as acoustic prompts. The TTS model then synthesizes isolated, speaker-specific “pure laugh” and “pure cough” tracks. To guarantee the acoustic fidelity and purity of these events, we apply a pre-trained SED model, PANNs [3], as an automated quality filter to eliminate samples whose detection confidence score is low. Subsequently, we process all retained clips through an energy-based VAD module to trim the leading and trailing silence (or artificial pauses) generated by the TTS engine, yielding tightly-bounded paralinguistic sounds. These synthetic clips are then ready for time-aligned mixing and frame-level annotation.

(3) Mixed audio generation: Guided by the pre-generated audio script, the validated paralinguistic clips are mixed with concatenated LibriSpeech utterances to form a meeting-style audio session. Following [10], the background events include nine classes of audio events from the DESED [17] soundbank. This rendering process faithfully executes the multi-talker overlap protocol defined by the script, seamlessly aggregating all isolated audio streams into complex acoustic mixtures with well-aligned ground-truth labels.

(4) Label Generation: Because the entire multi-stream mixture is constructed automatically through the proposed mixing engine, we bypass the pervasive issue of human annotation error or forced-alignment inaccuracies. The exact start and end timestamps of every speech segment, paralinguistic burst, and background event are deterministically aggregated to form frame-level ground-truth labels.

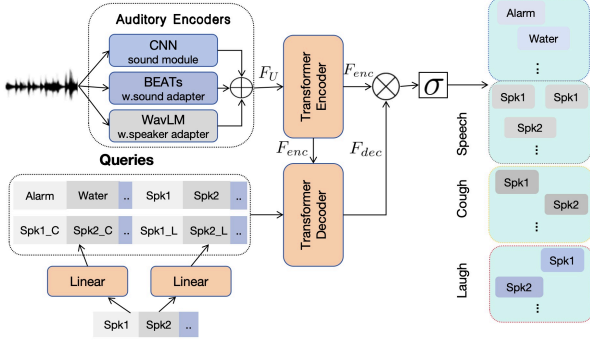


Figure 2: *Architecture of the proposed SA-UAED for Speaker-Attributed Paralinguistics.*

Built upon the above pipeline, we create a novel multi-label multi-speaker dataset, LibriPara by combining clean speech from Librispeech, synthesized paralinguistic clips, and background audios from DESED. It contains 500 hours of audios with multiple speakers at an overlap ratio of 15%, speaker-specific paralinguistic events, various background events, and fine-grained frame-level labels. This large-scale dataset can serve as a benchmark for developing speaker-attributed UAED systems. The audio simulation pipeline and dataset will be open-sourced at <https://github.com/originallover/SA-UAED>.

3.2. SA-UAED Architecture

Our proposed SA-UAED model is designed as a novel architectural paradigm for unified audio understanding. It builds upon the foundational encoder-decoder structure of the T-UAED [10] model, which was originally designed to unify SED and SD in a single model. The auditory encoder utilizes pre-trained self-supervised learning (SSL) models such as BEATs and WavLM [18] to extract rich audio representations. The Transformer-based decoder takes as input a series of learnable queries (corresponding to different types of audio events) and speaker embeddings to autoregressively generate frame-level activities of different sound events. While effective for its original tasks, the generic speaker embeddings in T-UAED are sub-optimal for modeling transient, non-verbal paralinguistic sounds such as laughter and coughing. As shown later in our experiments, the direct use of speaker embeddings for speaker-attributed paralinguistic event detection is problematic, showing poor F1 scores across two test sets.

To address this limitation, our proposed SA-UAED incorporates dedicated, task-specific fully-connected (FC) layers to project the generic speaker embeddings into distinct paralinguistic sub-spaces. The outputs of these dedicated FC layers are then used as additional paralinguistic queries in the Transformer decoder, operating in parallel with the original speaker embeddings and the learnable queries for audio events. The rest of the architecture (encoder, cross-attention, dot-product prediction, and binary cross-entropy loss) remains identical to the original T-UAED, ensuring seamless multi-task learning and minimal additional parameters. This design enables a single end-to-end model to simultaneously predict sound events, speaker activities, and speaker-attributed paralinguistic events at 50 Hz frame resolution.

4. Experimental Setup

All experiments are conducted on our proposed LibriPara dataset, comprising a 500-hour training set and a held-out 5-hour simulated test set generated via the pipeline described in Section 3.1. All audio clips are standardized to 32 seconds at a 16 kHz sampling rate, accompanied by frame-level annotations at a 50 Hz resolution. To closely simulate realistic conversational dynamics, each clip contains up to 3 distinct speakers, with the audio script generator enforcing a rigorously controlled average speech overlap ratio of 15%. During the mixing phase, 9 classes of background sound events from the DESED soundbank are injected at a Signal-to-Noise Ratio (SNR) ranging from 5 dB to 15 dB. Conversely, speaker-specific paralinguistics (generated via ChatterBox-Turbo with LibriSpeech references) are mixed at a tighter SNR range of 3 dB to 8 dB to preserve their impulsive acoustic characteristics.

4.1. Baselines

To ensure a fair comparison, we reproduce the original Transformer-based Unified Audio Event Detection (T-UAED) framework from scratch on our identical simulated dataset, using the exact same data splits, mixing parameters, and frame-level labels. The original T-UAED serves as the strong baseline, treating laughter and coughing as part of the general “speech”. To the best of our knowledge, SA-UAED is the first unified end-to-end framework tackling speaker-attributed paralinguistic event detection. While conventional cascade systems exist, they inherently suffer from severe error propagation in highly overlapped multi-talker scenarios. Therefore, we deliberately select the state-of-the-art unified architecture, T-UAED, as our primary baseline to isolate and demonstrate the impact of our proposed representation decoupling strategy.

4.2. Evaluation Metrics

To comprehensively assess the systems’ performance across diverse acoustic tasks, we adopt standard evaluation metrics tailored for high-resolution audio event detection and diarization.

For the detection of background sound events and speaker-attributed paralinguistic events, we report both the Intersection-Based F1 (IB-F1) [19] for event-level macro-detection and the Segment-Based F1 (SB-F1) [20], which are available at https://github.com/fgnt/sed_scores_eval/. Since our proposed simulation pipeline can directly provide frame-level ground-truth labels, we compute the SB-F1 metric evaluated at a high temporal resolution (matched to our 50 Hz / 20 ms frame rate) as our primary metric. This allows strict and precise quantification of the model’s frame-by-frame tracking capability in dense, overlapping acoustic scenarios. Furthermore, to evaluate the systems’ ability of speaker diarization, we report the standard Diarization Error Rate (DER), which comprises the summation of Miss (MS), False Alarm (FA), and Speaker Confusion (SC) errors.

4.3. Implementation Details

Our proposed SA-UAED and the original T-UAED baseline are both implemented in PyTorch. We use the Adam optimizer with an initial learning rate of 10^{-4} , which is decreased by 5% after each epoch. The feature dimension D is set to 192 for both audio features and UAED Queries. Specifically, the generic speaker embeddings acting as initial queries are extracted using a pre-trained ECAPA-TDNN model [21]. Following [10], the Transformer encoder and decoder each consist of 6 layers

Table 1: Comparison of IB-F1 and SB-F1 scores (%) on the LibriPara test set.

Model	IB-F1 (SED)	IB-F1 (cough)	IB-F1 (laugh)	SB-F1 (SED)	SB-F1 (cough)	SB-F1 (laugh)	DER (MS/FA/SC)
T-UAED Baseline	0.963	0.144	0.214	0.984	0.282	0.371	8.09 (2.54/1.68/3.88)
+ Shared Adapter	0.943	0.102	0.153	0.978	0.281	0.357	8.06 (2.63/1.58/3.84)
SA-UAED	0.942	0.248	0.230	0.979	0.473	0.476	8.07 (2.44/1.78/3.86)

Table 2: Comparison of IB-F1 and SB-F1 scores (%) on the EARS test set.

Model	IB-F1 (SED)	IB-F1 (cough)	IB-F1 (laugh)	SB-F1 (SED)	SB-F1 (cough)	SB-F1 (laugh)	DER (MS/FA/SC)
T-UAED Baseline	0.864	0.184	0.175	0.965	0.375	0.259	8.45(2.53/1.79/4.13)
+ Shared Adapter	0.829	0.147	0.134	0.959	0.369	0.256	8.51(2.63/1.49/4.39)
SA-UAED	0.825	0.192	0.220	0.962	0.470	0.409	8.43(2.47/1.90/4.39)

with 8 attention heads.

The dedicated fully-connected layers for cough and laughter queries are implemented as single linear layers ($192 \rightarrow 192$), adding only negligible parameters compared to the original T-UAED. Training is performed end-to-end with the binary cross-entropy loss as defined in Equation 1, where $n \in [1, N]$, $t \in [1, T]$, $\hat{y}_{n,t}$ and $y_{n,t}$ represent the predicted and ground truth labels of the n^{th} event for the t^{th} audio frame, respectively. The loss function is to minimize the difference between predictions and ground truth labels:

$$\mathcal{L} = -\frac{1}{N * T} \sum_{n=1}^N \sum_{t=1}^T (y_{n,t} \cdot \log \hat{y}_{n,t} + (1 - y_{n,t}) \cdot \log(1 - \hat{y}_{n,t})) \quad (1)$$

5. Results and Analysis

To rigorously evaluate our framework and its generalizability, we train all models from scratch solely on LibriPara (our 500-hour simulated dataset). The evaluation is systematically divided into two phases: assessing foundational multi-task learning capabilities on a held-out simulated test set (Table 1), and verifying zero-shot generalization on a real-world test set (Table 2) derived from the EARS [22] corpus, which provides both speech and paralinguistic audio clips from 107 speakers.

5.1. Performance on the Simulated Benchmark

As shown in Table 1, while the Baseline demonstrates competence in generic sound event detection (SED with SB-F1 of 0.984), it exhibits a catastrophic performance drop when identifying speaker-attributed paralinguistics (SB-F1 of 0.371 for laughter and 0.282 for cough). Notably, introducing a Shared Adapter does not resolve this, it severely degrades paralinguistic accuracy below the baseline. Forcing mutually exclusive acoustic events (rhythmic laughter vs. explosive coughing) into a shared projection subspace causes severe representation entanglement, thus greatly degrading the performance.

In contrast, our proposed SA-UAED model significantly circumvents this bottleneck by routing identity features into distinct, decoupled event sub-spaces. This task-specific adaptation achieves massive absolute improvements across both event-level (IB-F1) and frame-level (SB-F1) metrics, boosting the SB-F1 of laughter to 0.476 and coughing to 0.473, without noticeably impacting the DER.

5.2. Generalization to Real-World EARS Mixtures

To ensure the model does not merely overfit to TTS artifacts, we benchmark the models on the EARS [22]-based test set (Table 2), which is simulated following the same procedure as described in Section 3.1. To strictly isolate the evaluation of the paralinguistic sub-spaces from potential frontend speaker enrollment errors, we extracted the reference speaker embeddings via a pre-trained ECAPA-TDNN model [21], using a 5-second clean, non-overlapping speech segment from each target speaker prior to the mixture inference.

Table 2 provides compelling evidence of SA-UAED’s zero-shot generalization to genuine human vocalizations across both evaluation resolutions. On the macro level, the consistent gains in IB-F1 confirm that our decoupling strategy successfully transfers to spontaneous human behaviors, robustly capturing “in-the-wild” paralinguistic events. More impressively, on the micro level, the SB-F1 for EARS-based laughter detection leaps from 0.259 (Baseline) to 0.409, and cough from 0.375 to 0.470. This demonstrates that even when facing highly irregular, real-world human vocal bursts, SA-UAED maintains exceptional temporal boundary fidelity, proving the sheer robustness of the proposed task-specific adaptation.

Both tables reveal an intriguing trade-off: the substantial gains in paralinguistic detection come at the cost of a marginal degradation in generic SED performance (e.g. dropping slightly from 0.984 to 0.979 on simulated data). This phenomenon highlights a classic challenge of negative transfer. By forcing the shared acoustic frontend to simultaneously optimize for continuous articulated speech, environmental sounds, and transient paralinguistic bursts, the shared representation space inherently shifts. Future unified audio frameworks may need to explore decoupled audio encoders to eliminate representation interference at the feature extraction stage.

6. Conclusion

This paper addresses the limitations of existing unified audio systems in speaker-attributed paralinguistic detection. We propose a highly controlled simulation pipeline for generating complex acoustic mixtures with precise frame-level multi-label annotations, and use it to construct LibriPara, a 500-hour multi-talker dataset integrating continuous speech, background sounds, and speaker-specific paralinguistics. We further introduce SA-UAED for joint frame-level detection of audio events, speaker activities, and speaker-attributed paralinguistic events. Future work will explore self-supervised representation alignment and open-set vocalization detection.

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8. Generative AI Use Disclosure

We acknowledge that AI tools based on large language models were used for grammar correction and description polishing. All AI-generated sentences have been manually checked and refined to ensure correctness and preciseness.

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